Supplementary Information:
The Lonely Runner Problem on Pair-Sum Lattices

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1 The verification record of the scaling-closure derivation

Every number above is either an exhaustive census with two independent decision procedures, or a sample with its size and interval, or a proved statement with its machine check; the records, in the order of appearance: the pool censuses (the pruned-cascade and the flattened-popcount procedures agree exactly at $p = 17/19/23$; an independent pure-enumeration brute force agrees on 21/21 sampled tuples across $p = 17/19/23/31$, covering both admissible and inadmissible branches; the $T = 9$ stride samples carry full decisions, and the $m = 6$ code path shares the cascade code brute-force-verified at four $m = 5$ primes, with the external consistency check the agreement of the projected pools with the independently measured pool scale); the zone-resolved γ table (764 pairs, five committed cells, the identical seed reproducing the committed family selections); the chain-class machine checks (123,240 solutions, 0 violations of the sum identity or the reach bound; the count reconciliations 131,520 against 123,240 and 10,080 against 8,880); the reflection fiber (survival exactly 1 at every $\bar{\lambda} = 1$ pair, every family, every cell); the renormalization-orbit law (machine-checked set-by-set at $p = 17$ and $p = 19$); the volume census and its ground truth (9/9 brute-force keys; the admissibility reproductions 840/840 and 960/1,680; the order-independence of volume across all 24 orderings); the essential-arc, budget-identity, and cap checks on the probed families; and the cascade depths at the committed cells (medians 2, maximum 4); and the character-system record of §3.5 (the rigidity census over 1,678 admissible keys with every key inside the committed census scope matching it exactly, 1,654/1,654, 0 mismatches; the per-assignment renormalization bijection, 801/801 images injective with per-key counts matching under the explicit σ; the falsifications of V' and V''; the character-system aggregate invariance on both full top orbits; the chain controls 8,880/4,680/29,760 reproduced by the same code path). The transfer-matrix route's record: the generic dichotomy and trace fingerprints (1,500 systems, 12,000 comparisons, zero mismatches); the 66 homogenized pair systems on the six committed families with the closed-form orbit inventories verified by full traversal and the finite-horizon survival counts verified by direct simulation (zero mismatches); the loop-composition identity (algebraic and numeric, 500 points per family); the per-step equality of the automaton's survivor counts with the committed μ-space cascade convention (zero mismatches, all steps, all families; the committed chain-family depth reproduced); the end-to-end fiber-covering ground truth (2,800/2,800); the reflection pairs (survival fraction exactly 1, all six families, all three primes); and the reproduction of the committed volume margins (876/108/186 against $95\times$ /1,207 $\times$ /3,803$\times$). The three-rung depth record of Table 3 is calibration from the earlier cascade runs and is labeled as such wherever it is used.

2 Reproducibility and soundness record

Every empirical and classification claim in this paper rests on exact integer computation, and every computation is paired with a control or an independent reimplementation. This section records the checks; the committed scripts, logs, and experiment notes are bundled with the paper [17]. The record matters beyond convention here: a measurement program on 10^6-scale corpora is exactly the setting where silent bugs masquerade as findings, and the discipline below is what separates the two.

Table 1: The verification record. All counts are of exact checks with the stated outcome; "0 mismatches" means exact agreement of two independent computations.

check	scope	outcome
size law (Lem. 4.1)	all residues, $N \le 111$ (T=6), $N \le 31$ (T=5)	0 violations
classification predicate $\iff$ mask covering	exhaustive over unit multisets at $7, 11, 13, 17, 19, 37$ (126–82,251 per modulus)	0 mismatches
K_{19}/K_{37} class list $\iff$ covering	all unit 4-multisets, both directions (5,985 and 82,251)	0 mismatches

Table 1 (continued)

check	scope	outcome
battery soundness (flag ⇒ certify)	exhaustive $N \le 31$ (46,371 at n=4; 324,626 at n=5); exhaustive $N \in \{37, 49, 81, 95\}$ (91,390; 270,725; 1,929,501; 3,612,280); exhaustive $N \in \{99, 111\}$ (4,249,575; 6,672,876); random 32–95 (256,000); random 96–111 (64,000)	0 violations
polygon constructive witness	sampled across all corpora	0 failures
reduced-optimum cross-check	direct grid maximization vs. bitmask, sampled	0 mismatches
lift law (Lem. 4.5)	random lifts	0/2,727
equality theorem (Thm. 3.2)	fresh non-circular check (folklore breakpoints vs. pair lattices), 885/885; from-scratch independent reimplementation (time domain, differences included, $n = 2$–6, $v \le 200$), with exact anchors	2,870/2,870
ground truth of (TAU-$_5$)	every primitive 5-set, $v \le 56$	3,712,576/3,712,576
trichotomy proof, line by line	every intermediate claim of Lemma T (non-wrapping, span bounds, two-lattice descent, two-block identity) at all primes $p \le 250$, plus the run lemmas at $p \le 37$	0 violations
colliding-unit ground truth	all ±-colliding $(1K, 3U)$ families, both collision topologies, $p \le 43$	3,3 M families, 0 coverings
shear linearity	per-fiber t-space offsets of colliding lifts, all v_2, m, j at $p \le 37$	0 violations of $\sigma_3 - \sigma_2 = -\bar{u}mj$
boundary finite case analyses	Lemma B at 11; Lemma A at 19: every viable configuration certified, independent brute force in agreement (and reproducing the known failures 50 at 7, 2 at 13 as witness-tuple counts; the $17, 31, 37$ analyses are redundant with the extended proof ranges)	exact
group-ring identity (Thm. 6.1)	36,308 checks: all 52 Lemma A exceptions (E-profiles exactly as classified), every covering of the census ($11 \le p \le 61$) plus boundary $p = 5, 7, 13$ (12,256), 24,000 random configurations incl. non-coverings (sharp signed form; sign ⇔ covering; kernel $= \mathbb{Z}\overline{\Phi}_p$)	all pass
covering census (Prop. 6.4)	all critical coverings over all $(d_2, d_3) \in (\mathbb{Z}_p^*)^2$, normal form, $p \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61\}$	four families, counts constant in p; 0 ±-distinct coverings at $p \ge 11$
foot census (Input E)	every prime $p \le 419$, both classes, every critical (s_1, s), ±-reps $d \ge 2$: one/two-foot viable sets; the M4/M5 consumed pairs; Lemma B's contained census at $(6, 6)$	consumed form clean from $p \ge 31$; now *proved* (Lemma 5.16)
Lemma E machinery (Lemmas 5.11–5.16)	every (p, d, s) with $p \le 500$ (45 primes, 10,410 triples): column decomposition, band and block identities, the pair-sum identity as an *exact* equality, the auxiliary bounds; the theorem direct to $p \le 2000$ (136,806) and, via the identity, to $p \le 4000$ (367,986); endpoint/dirty-prime values exact; 76/76 viable-set agreement with the foot census	all hold
parity mechanism (Prop. 6.5)	every census $(2, 2)$ solution at $p \in \{23, 29, 41, 31, 37, 43\}$: integer-parity runs, non-wrapping, spills ≤ ov in A_1; per-(multiset, family) counts	100%; counts = Table 8
pq translation (Thm. 7.1)	constants; footprint formula (4,394 set equalities, 8 moduli); row census (24 resp. 312, 14 primes); $(1, 1, 2)$ cells (144,138 checks, 11 moduli 35–143, including $N = 133$); $(1K, 3U)$ cells (989,000 checks); pinning (37,672 rows, ≤ 6 good rows); T-10 reproduction at 77, 91	all pass; 0 coverings
higher-rung machinery (Lem. 8.1)	three-tier arc law vs the committed $T = 6$ tables ($p \le 37$, both classes) and at 19 primes of $T = 7/T = 8$ (all residue classes, every a_0); footprint formula vs direct Z_{p^2} enumeration; drift identity at $T \in \{7, 8\}$	957/957; 3,528/3,528
$T = 8$-critical census (Prop. 8.2)	17 primes, all four odd ϵ classes, $k \le 17$: exactly 11 families, counts 288 ($\epsilon \in \{3, 7\}$) resp. 2,784 ($\epsilon \in \{1, 5\}$) constant in p; 0 ±-distinct in the mechanism range	exact

Table 1 (continued)

check	scope	outcome
$T = 7$ census (Prop. 8.3)	18 primes, all six ϵ classes, $k \leq 11$: core-eleven dominance ($\approx 90\%$ at $k \geq 6$), tail decay $183 \to 43$, 0 $\pm$-distinct at $k \geq 4/k \geq 2$	exact
higher-rung cells (Prop. 8.4)	$(1K, 4U)$ at $T = 7$: census-restricted ground truth (distinctness-filtered) at every prime $11 \leq p \leq 61$ — 0 coverings, best exit 2; brute-force control at $p \in \{5, 7, 11\}$ agreeing exactly at the degenerate $p = 5$ (8 coverings both ways); $(2K, 4U)$ at $T = 8$ at $17 \leq p \leq 43$ — 0 coverings; sampled controls 20K families at four primes	exact
$n = 4$ control	full record incl. the open set $= (3, 5, 8, 13)$	exact
range controls, both batteries	$V = 16/24/32/40/48$, all tiers, unflagged pair totals and decompositions	exact vs. committed logs
capability controls	zones, counts 20/68/16/64/32, safety of 48–111	exact
independent capability recounts	set-based scanner at all danger moduli (27 moduli)	exact agreement

2.1 Controls

Every experiment run begins with hard assertions against previously committed numbers before any new number is produced: the $n = 4$ corpus reproduces the committed record (including the identity of the open set); the $n = 5$ corpora at $V = 16, 24, 32, 40, 48$ reproduce every committed tier count for both batteries, the corpus-wide unflagged-pair totals and decompositions included; the capability scan reproduces the committed zones, the per-modulus covering counts, and the safety of 48–111 (the composite range extended twice, both times with the new moduli re-derived by the independent set-based scanner); and the signature-capability maps for 48–79 and 80–95 match the committed dictionaries exactly. The classifications were additionally anchored by hand: the $\mathbb{Z}_{17}$ covering of Example 5.6 was expanded by hand, the D_3 falsifier $(1, 1, 2, 3)$ on $\mathbb{Z}_7$ and its exclusion $N \equiv 1 \pmod 6$ were checked by hand, and the K_{19}/K_{37} covering counts were reproduced by three independent implementations (bitmask, frozenset, exponent-space).

2.2 The cross-validation discipline

The program's standing rule is that every empirical claim must be checked against committed data or a reference implementation before being cited. The rule earned its keep: across the program's lifetime it caught, before any affected number was recorded, a divisibility test written as a bitwise OR; a sampling artifact in which a 720-point time grid silently missed denominators like 11 and briefly masqueraded as counterexamples to the equality theorem; an unsound counting bound for mixed-order residues (245 flagged-but-non-certifying pairs, all caught by the flag-implies-certify sweep); an off-by-one in the capability scanner that contradicted the committed $n = 4$ zone; a flag-parsing slip; a verified-tier flag leaking into a proved tier (caught by a too-good "100%"); a diagnostic accumulating corpus-wide instead of per-tier (caught by an impossible count); a mislabeled canonicalization of cyclic translate classes (contains-0 representatives are not orbit representatives on $\mathbb{Z}_m$); and an unsound attempt to quotient the classification by exponent inversion (class reflection $e \mapsto -e$, which fixes the bad set only through $B_{-w} = B_w$ and is not a covering symmetry); an off-by-one in the minimal-arc length of the census fit test (caught by the assertion that every enumerated covering has $E \geq 0$, fixed, and then confirmed against a no-prune brute force); and a count-semantics reconciliation forced by the census run (the earlier 50/2 at $p = 7/13$ are witness-tuple counts, one per (d_2, d_3, a_2) with a completing placement, while the true solution counts are 272 normalized configurations $= 68$ set-triples and $8 = 2$ — the brute force confirms both readings, and the identity check ran on the full solution set). Each fix was verified by the control that caught it. We report these not as penance but as evidence that the numbers above survived an adversarial process.

2.3 Artifacts

The bundle [17] contains the equality-theorem verifications (`V1_equality_fresh_check.py` and the from-scratch `EQ_independent_verify.py`); the independent capability cross-checks (`N5_prime_scan_crosscheck.py` and the set-based scanners); the literature-search scripts and raw results; the experiment notes summarizing each stage; and the corpus and battery engines with their controls, one per stability extension, each with its full run log:

- `N5_leverage_experiment.py` (the committed $n = 5$ battery and rigid-zone scan),
- `N6_stability_experiment.py` (the $V = 32/40$ extension and the composite capability completion to $N \leq 79$),
- `N7_classify_19_37.py` (the K_{19}/K_{37} classifications and the augmented re-measurement),
- `N8_v48_extension.py` (the $V = 48$ data point, both batteries, capability to $N \leq 95$),
- `N9_v56_extension.py` (the $V = 56$ data point, both batteries, capability to $N \leq 111$; staged execution with per-stage state persistence and hard controls re-asserted at every stage),
- `ap_lemmas_verify.py` (the trichotomy/Lemma A/B verification of the companion note: exact lemma search, censuses, capacity structure, cell anchors; $p \leq 250$),
- `trichotomy_linecheck.py` (the line-by-line machine check of the trichotomy proof, including the general-arc variant),
- `pm_colliding_close.py` (the $\pm$-colliding closure: ground truth, fiber structure, shear linearity, pinning tables, pigeonhole),
- `boundary_finite_cases.py` (the boundary-prime finite case analyses with margin certificates and the independent brute-force cross-check),
- `groupring_bridge.py` (the group-ring translation verification: identity checks on all exceptions and census coverings, sharp-form sampling, kernel checks, the four-family census with per-multiset counts, defect statistics, and the witness/solution count reconciliation; output `out_groupring_bridge.json`, log `groupring_run.log`),
- `foot_census.py` (the Input E boundary census: one/two-foot viable sets at every prime $p \leq 419$, the consumed-pair cleanliness, and the Lemma B $p = 17$ form check; output `out_foot_census.json`),
- `lemmaE_master.py` (the three-distance foot-count verification, discharging Input E: column decomposition, band and block identities, the pair-sum identity as an exact equality, the auxiliary bounds of Lemma 5.15, the theorem of Lemma 5.16 direct to $p \leq 2000$ and via the identity to $p \leq 4000$, the endpoint and dirty-prime values of Remark 5.17, and the viable-set cross-check against the foot census; output `out_lemmaE.json`),
- `pq_side.py` / `pq_side2.py` / `parity_check.py` (the pq translation battery: constants and footprint formula, row censuses, $(1, 1, 2)$ and $(1K, 3U)$ cell enumeration with margins, pinning and drift analysis, and the parity-mechanism structural check; outputs `out_pq_side.json`, `out_pq_side2.json`),
- `higher_rung_step1.py` with the resumable `drive_step1.py` (the general-T machinery controls and the higher-rung censuses: the $T = 7$ four-difference census at 18 primes with per-family projections, the $T = 8$-critical census at 17 primes, the $T = 8$ five-difference probe; output `out_higher_rung_step1.json`, log `higher_rung_run1.log`),
- `higher_rung_step2.py` (the shear side at the higher rungs: the drift-law verification, the census-restricted and brute-force ground truths for the $(1K, 4U)$ cell at $T = 7$ and the $(2K, 4U)$ cell at $T = 8$, the exit-time measurements; output `out_higher_rung_step2.json`),

- `higher_rung_step3.py` (the sampled family controls for the two frontier cells; output `out_higher_rung_step3.json`),
- `pool_sat.c` with `gt_pool_brute.py` (the pool saturation censuses of §3.2: the m-general admissibility checker in two cross-validated implementations plus dump/resume stride sampling, and the independent pure-enumeration brute force agreeing on 21/21 sampled tuples across $p = 17/19/23/31$; dumps `dump_p{17,19,23}.tsv`, `pool_T8_p31.tsv`, `dump_T9_p{29,31}_sample.tsv`),
- `h2_pair_check.py` (the zone-resolved cut-quality measurement of Table 4: per-pair γ on the committed cells, 764 pairs; output `out_h2_pair_check.json`),
- `chain_structure_check.py` (the chain-class machine checks of Lemmas 9.7–9.8 and Theorem 9.9: 123,240 solutions, the sum identity, the reach bound, and the count reconciliations),
- `scaling_closure_extract.py` and `scaling_closure_verify.py` (the read-only calibration from the committed JSONs and the targeted lemma verification of §3.3: solution-volume probes, two-fiber dilate law, resonance zones, cascade depths at the committed cells, sign robustness; outputs `out_scaling_closure_extract.json`, `out_scaling_closure_verify.json`),
- `h1_t8_census.py`, `h1_t8_structure.py`, `h1_t8_prime.py`, `h1_t8_path.py` (the $T = 8$ volume census, mechanism probes, and prime extensions of §3.5, with the independent brute-force ground truth (9/9 keys) and the renormalization-orbit verification; states `h1_t8_state.json` / `h1_t8_prime_state.json`, outputs `out_h1_t8_census.json`, `out_h1_t8_structure.json`, `out_h1_t8_prime.json`).

All runs are deterministic; every number in §10–2 appears verbatim in a committed log.

2.4 Data and code availability

The complete artifact bundle (scripts, committed logs, JSON records, and the figures) accompanies this paper and is deposited for public access together with the PDF; the sources build with `tectonic` and a single command. The verification scripts have no dependencies beyond CPython 3.12 with `numpy`; each re-derives its controls from first principles and asserts them before producing any new number, so any single experiment can be re-verified in isolation. This document and the bundle are released under the Creative Commons Attribution 4.0 license (CC BY 4.0), the better to allow the community to re-run, refute, or extend the record.

3 The scaling-closure lemma: derivation and calibration

This section is the derivation record behind Theorem 2.1. Its order is the order of the reduction: the cascade model and the dilate-intersection step (§3.1); the pool law, its saturation tables, and the status of the 840 identity (§3.2); the cut law—spacing, k-cancellation, and the conditional comparison that fixes the depth formula's two ends (§3.3); the two-regime structure of the dilate cuts, with the reflection fiber and the chain class proved (§3.4); the volume law at $T = 8$ with its threshold form, the renormalization lemma, the character system, the reduction of the remaining obligation to the strand census, and the transfer-matrix route that closed the third route by theorem (§3.5); and the epistemic status (§9.1). Throughout, a claim is labeled *proved* (with its verification record), *measured* (with its scope, sample size, and, where sampled, an interval), or *fit* (with the data it is fit to); extrapolations beyond the tested range are labeled *projected*; and no claim is made about the amount of work needed to close the open obligations.

3.1 The model: the cascade as a dilate-intersection

Fix the rung $T \geq 8$, the prime $p = Tk + \epsilon$, and the composite modulus $N = p^2$. The frontier cell $(1K, mU)$, $m = T - 3$, consists of the covering families with one kernel member and m unit members; by Lemma 8.1 the kernel covers the fibers of a kernel bad set B_k of size $2k + 1$, leaving $|U| = (T - 2)k + \epsilon - 1$ uncovered fibers $F_j \cong \mathbb{Z}_p$, and on each fiber the m unit footprints are arithmetic progressions whose sizes lie in the two-value window and whose starts obey the drift law. In the census normal form—the normalizer arc pinned with difference 1 at the origin, the remaining differences canonical in $[2, D]$, $D = (p - 1)/2$—the placement on fiber j is the relative-start tuple

$$c(j) \;=\; c^0(j) - \mu\, j \;\in\; (\mathbb{Z}_p)^{m-1},$$

where $c^0(j)$ is the residue-determined part (a function of the unit residues and the three-tier size assignment $\mathrm{sz}(j)$) and $\mu \in (\mathbb{Z}_p)^{m-1}$ is the free drift, one coordinate per moving arc, determined by the lifts. The fiber is covered if and only if $c(j) \in \mathrm{Sol}(\mathrm{sz}(j))$, the census solution set of the size tuple and the family's difference tuple; this is exactly the committed ground-truth check. The *fiber cascade* processes the uncovered fibers one by one, retaining the anchor placements that survive each; the family dies when the survivor set empties, and the *cascade depth* is the number of fibers needed.

Lemma S. 3.1 (two-fiber reduction)**.** *Fix an anchor fiber $j_0 \in U$ and an anchor solution $a \in \mathrm{Sol}(\mathrm{sz}(j_0))$. The anchor determines the drift, $\mu = (c^0(j_0) - a)\, j_0^{-1}$. For a second fiber j with $\lambda := j\, j_0^{-1} \bmod p$, the anchor solution a survives fiber j if and only if*

$$\lambda\, a + \tau \;\in\; \mathrm{Sol}(\mathrm{sz}(j)), \qquad \tau = c^0(j) - \lambda\, c^0(j_0) \;\in\; (\mathbb{Z}_p)^{m-1}.$$

Hence the two-fiber candidate set is the dilate-intersection $\mathrm{Sol}_0 \cap \lambda^{-1}(\mathrm{Sol}_j - \tau)$, and the cascade after i processed fibers is the iterated intersection $\bigcap_{\ell \leq i} \lambda_\ell^{-1}(\mathrm{Sol}_\ell - \tau_\ell)$.

Proof. Substitute $\mu = (c^0(j_0) - a)\, j_0^{-1}$ into the covering condition $c^0(j) - \mu\, j \in \mathrm{Sol}(\mathrm{sz}(j))$ and rearrange; all maps are affine over $\mathbb{Z}_p$. □

The cascade step is thus exactly a *dilate-intersection of census solution sets*: the covering constraints across fibers are affine conditions on the anchor placement, with dilation ratios $\lambda = j/j_0$ given by the fiber set. This is the precise form of the picture under which the anchored-core analysis was organized. Write $\bar{\lambda} = \min(\lambda, p - \lambda)$ for the cyclic representative. The dilate $\nu \mapsto \lambda\nu$ maps the junction range into itself strongly when λ is a *small rational* ($\bar{\lambda} \leq \Lambda_0$, or $\lambda \equiv a/b$ with $a, b \leq O(L)$ for the junction length L): these are the *resonant* fibers, where the cut is weak. The remaining fibers impose genuine spacing (Lemma 3.3). At the committed zoo prime ($p = 17$, $k = 2$) the resonance zone $\bar{\lambda} \leq 4$ covers half the dilations—the small-p extreme; its fraction vanishes as p grows. The per-pair *cut quality* is measured by

$$\gamma \;=\; \frac{\#\big(\mathrm{Sol}_0 \cap \lambda^{-1}(\mathrm{Sol}_j - \tau)\big)}{|\mathrm{Sol}_0|\,|\mathrm{Sol}_j|\,/p^{m-1}},$$

the survivor count against the random-placement baseline; $\gamma = 0$ is an *instant kill* (the actual intersection is smaller than the random model), $\gamma \approx 1$ is no cut at all, and $\gamma \gg 1$ marks a resonant pair whose solution sets are too small for the random baseline to be meaningful.

The ±-reduced convention. The cascade is modelled on a ±-representative fiber set $U_\pm$: one representative per reflection pair $\{j, -j\}$, the reflection fiber excluded. This is forced by Lemma 3.6 (the bad sets are symmetric under $x \mapsto -x$, so fiber $-j$ imposes exactly the same covering condition as fiber j); processing both would double-count every constraint, halve the effective fiber budget, and make the survivor inequality $\mathrm{Hq}_\pm$ fail on the reflection duplicate. The measured depths of Table 3 are oriented-fiber counts (both j and $-j$), exceeding the ±-reduced formula by this duplicate; the theorem's budget is $|U_\pm| = \frac{1}{2}|U|$.

3.2 The pool law

Lemma S. 3.2 (pool law)**.** *The family pool after the ±-collision reduction is bounded by the distinct canonical difference tuples:*

$$P(T,p) \;\le\; (D-1)(D-2)\cdots(D-(T-4)) \;\le\; \big((p-1)/2\big)^{T-4}, \qquad D=(p-1)/2,$$

with difference count $T-4$ (the normalizer's difference is pinned to 1).

Proof. The census key of a ±-distinct family is an ordered tuple of distinct canonical differences in $[2,D] \subset [1,D]$; the falling factorial counts them. □

The law that is *not* a theorem is the saturation identity $P=(D-1)_{T-4}$. The committed zoo point ($T=8$, $(1K,5U)$, $p=17$, $k=2$) records exactly 840 ±-distinct families and $840 = 7{\cdot}6{\cdot}5{\cdot}4$—the bound saturated, every distinct tuple admitting a covering placement. Table 2 is the verdict at the demanded primes, and it is unambiguous: *the identity is a local accident of the slackiest prime, and the pool law is rung-monotone.*

Table 2: Pool saturation at the frontier cells. The $T=8$ rows are exhaustive censuses (two independent decision procedures, cross-checked against a third brute force; §3.6). The $T=9$ rows are stride-sampled with full decisions per sampled tuple: saturation carries a 95% Wilson interval with the stated n; the projected pools (bound × sampled fraction, one prime each) are *projected*, not established.

rung	p	(k,ϵ)	bound $(D-1)_{T-4}$	admissible	saturation
$T=8$	17	$(2,1)$	840	840	1.000
$T=8$	19	$(2,3)$	1,680	960	0.571
$T=8$	23	$(2,7)$	5,040	1,800	0.357
$T=8$	29	$(3,5)$	17,160	2,280	0.133
$T=8$	31	$(3,7)$	24,024	360	0.015
$T=9$	29	$(3,2)$	154,440	—	0.903 $[0.889, 0.915]$
$T=9$	31	$(3,4)$	240,240	—	0.490 $[0.468, 0.511]$

Three readings. *First*, the saturation mechanism: a covering by arcs with pairwise-distinct differences must waste overlap—the chain mechanisms of the repeated-difference core all exploit pairing (the interval tiling needs $d=1$ twice, the parity mechanism needs $d=d'$, the antipodal needs the q-pairs), so saturation requires the overlap budget to absorb the *distinctness penalty*. The budget-versus-penalty race is lost at $T=7$ (the distinct pool is empty for $k\ge 2$: the committed 18-prime census records 24/24 at $p=11$ and 36/60 at $p=13$ at $k=1$, and zero at every $k\ge 2$ prime), marginal at $T=8$ (it holds only at $p=17$, the prime with the largest overlap window, $\mathrm{ov}\in[3,8]$; it collapses as the window tightens and as k grows), and won with margin at $T\ge 9$ (at $T=9$ the extra arc's budget $\mathrm{ov}\approx (T-6)k+O(1)$ covers the penalty for most tuples: 90.3% at the slackier $\epsilon=2$ window, 49.0% at the tighter $\epsilon=4$). *Second*, the projected $T=9$ pools: 1.39×10^5 (interval $[1.37,1.41]\times 10^5$) at $p=29$ and 1.18×10^5 ($[1.12,1.23]\times 10^5$) at $p=31$—the measured $\sim 10^5$ pool scale at $T=9$ is (approximately) the distinct-tuple pool, and the $T=10$ reading ($\sim 10^8$ at $k\ge 4$, the falling factorial times the anchor multiplicity) is consistent with the rung-monotone law; the $T=10$ saturation itself (predicted ≥ 0.9 at the 40% budget) is *not measured*, and at $k\ge 4$ the pools exceed the cascade's reach in any case. *Third*, the set-form: admissibility is invariant under permuting the differences (0 order-conflicts over the 6,720 checked tuples at $p=19$ and $p=23$, forced by the size-assignment symmetry of the census), so the pool is $\#\mathrm{sets}\times(m-1)!$ with $\#\mathrm{sets}\le\binom{D-1}{m-1}$, and the saturation statement becomes "all $\binom{D-1}{m-1}$ sets admissible"—true only at the slackiest (rung, prime) pairs, asymptotically in T. The absolute admissible count at $T=8$ grows across the table ($840\to 960\to 1{,}800\to 2{,}280$) but collapses at $p=31$ (360): the pool is real but ϵ-modulated, never a smooth polynomial in p.

3.3 The cut law: spacing, k-cancellation, and the comparison

Lemma S. 3.3 (dilate-count spacing bound)**.** *Let p be prime, $\lambda \in \mathbb{Z}_p^*$, $0 \le L < p$, $t \in \mathbb{Z}_p$. Then*

$$N(\lambda, t) \;:=\; \#\{\nu \in \{0, \dots, L\} : (\lambda\nu + t \bmod p) \in \{0, \dots, L\}\} \;\le\; \lfloor L/b_{\min} \rfloor + 1,$$

where $b_{\min}(\lambda, L) := \min\{b \ge 1 : \|\lambda b\|_p \le L\}$ is the first return of the λ-walk to the L-ball ($b_{\min} = L + 1$ if no such $b \le L$ exists, giving $N \le 1$).

Proof. Let $\nu_1 < \nu_2$ be two solutions. Then $\delta = \nu_2 - \nu_1 \in [1, L]$ satisfies $\lambda\delta \equiv (\lambda\nu_2 + t) - (\lambda\nu_1 + t)$ with both right-hand residues in $[0, L]$, so $\|\lambda\delta\|_p \le L$; hence $\delta \ge b_{\min}$. The solutions are therefore $b_{\min}$-separated in ν, and at most $\lfloor L/b_{\min} \rfloor + 1$ of them fit in $[0, L]$. □

The corollary is the resonant/non-resonant split. The survival fraction per coordinate is $\sigma(\lambda) := \max_t N(\lambda, t)/(L+1) \le 1/b_{\min} + 1/(L+1)$: for an integer dilate $\bar{\lambda} \in [2, L]$ one has $b_{\min} = 1$ and $N \approx (L+1)/\bar{\lambda}$, so the cut is weak for $\bar{\lambda} \le 2$ and real for $\bar{\lambda} \ge 3$; for a small rational $\lambda \equiv a/b$ with $a, b \le O(L)$ one has $b_{\min} \le b$ and the cut degrades by the resonance factor, the number of such dilations among the fibers being $\le \sum_{b \le B}(2L+1)$ for strength B—the resonance budget; and otherwise (no good small-rational representative —the bulk of the fibers at large p) $b_{\min} \gtrsim p/(2L+1)$, so $\sigma(\lambda) \lesssim (2L+1)/p + 1/(L+1)$, and with $L = Rk$, $p = Tk$: $\sigma \lesssim 2R/T$. One regime caveat must be recorded: the box form requires $L = Rk < p/2$, i.e. $R < T/2$—true for the thin majority of zoo families ($R \approx 4.8$ at $T = 8$) but *false* for the fat top families (R up to ≈ 10.5, k-free). For those, the cut is carried not by the spacing lemma but by the solution sets' own sparsity (their volume is far below p^{m-1}; the measured instant kills), which is exactly H2m's content.

Proposition S. 3.4 (k-cancellation)**.** *Assume the volume law in either proved or hypothesized form—for the chain class, $|\mathrm{Sol}| \approx \mathrm{ov}^{m-1}$ with $\mathrm{ov} \approx (T-6)k$ (Lemma 3.8); for general families under H1g, $|\mathrm{Sol}| \le c_1\,\mathrm{ov}^{m-1}$. Then the per-fiber cut ratio at the frontier cell is k-free:*

$$\text{cut} \;\approx\; \frac{p^{m-1}}{|\mathrm{Sol}|} \;\gtrsim\; \Big(\frac{T}{T-6}\Big)^{m-1} / (\textit{order and tie factors}).$$

Proof. The junction volume and the modulus scale together: $p^{m-1} = (Tk)^{m-1}$ against $|\mathrm{Sol}| \approx ((T-6)k)^{m-1}$, so the k^{m-1} in the denominator cancels the k^{m-1} in the numerator and the ratio is a function of T alone, up to the order factors (the $(m-1)!$ orders and the tie bookkeeping), which are k-free by the census rigidity. □

The earlier form of this proposition was stated with a fitted junction scale R ($|\mathrm{Sol}| \le (Rk)^{m-2}$, $R \approx 4.2$); the chain-class analysis of §3.4 falsified that exponent for the dominant family (the correct chain exponent is $m-1$, not $m-2$) and *derived* the volume constant from the overlap budget $T-6$. No fitted R is used; under H1g the thin families' constant is c_1, bounded by data. Either form leaves the cut k-free—the cancellation is robust to the correction.

Proposition S. 3.5 (comparison; the ±-reduced cascade, evaluated under the calibrated constants)**.** Derivation, not a cascade theorem*: the model below is the calibration of Theorem 2.1's clause (i) under Proposition 2.2's ±-reduced induction ($Hq_\pm$ with additive error term E, quantified over the ±-representative non-resonant fiber set $U_\pm$). Write $d = m - 1 = T - 4$, $B = c_1\mathrm{ov}^d$, and $r = \Gamma\,(p/\mathrm{ov})^{-d}$ (the per-fiber contraction factor of Prop. 2.2). The survivor count after q processed fibers obeys $|S_q| \le B\,r^q + \frac{E}{1-r}$, so the cascade reaches the $O(1)$ floor $|S| \le E/(1-r)$ within*

$$q \;=\; \frac{\log(c_1\,\mathrm{ov}^{m-1})}{(m-1)\log(p/\mathrm{ov}) - \log(c_1\Gamma)} \;+\; O(1)$$

fibers (the leading term; the $O(1)$ absorbs the error-term floor), after which a single further fiber kills the structured $O(1)$ remainder. The ±-representative fiber budget is $|U_\pm| =$

$\frac{1}{2}((T-2)k+\epsilon-1)$. *Consequently:* (a) *(depth is $O_T(\log k)$, never linear) the leading term is* $\dfrac{(m-1)\log \mathrm{ov}+O(1)}{(m-1)\log(p/\mathrm{ov})-\log(c_1\Gamma)}=\dfrac{\log k}{\log(T/(T-6))}(1+o(1))$*, which grows like* $\log k$ *with a T-dependent slope and has* no *k-uniform linear bound;* (b) *(T-flat plateau at practical k) the pool-route term stays in* $[1.1, 2.9]$ *for* $k \le 10$, $8 \le T \le 20$, *so the typical depth is* $\approx$ 2–3 *plus the* $O(1)$ *floor;* (c) *(the "$\le T+3$" form is not used) the quantity* $(T-4)+N_{\mathrm{res}}$ *is* not *an upper bound on the depth and is not k-uniform; the explicit $O_T(\log k)$ formula above is the depth bound;* (d) *(closure margin against the $\pm$-budget) for every* $T \ge 8$, $k \ge 2$ *the contraction count q satisfies* $q \ll |U_\pm| = \frac{1}{2}((T-2)k+\epsilon-1)$, *with surplus growing linearly in k; equivalently the per-fiber factor r may be as large as* $(T/(T-6))^{m-1}$ *(the threshold value) and the cascade still closes within the $\pm$-budget. The factor* $\frac{1}{2}$ *in* $|U_\pm|$ *is the cost of the reflection identification (Lemma 3.6: fibers $\pm j$ are one constraint).*

Table 3: The depth calibration at the three rungs. Measured: the committed cascade record (maxima over $k \le 9/8/3$; medians 2/3/5). Formula: Proposition 3.5 evaluated with derived constants ($c_1 = 1$, $\Gamma = 1$) at the top-of-window overlap; the window shifts the leading term by ± 1.5 at $T = 8$. The "worst case" column is *withdrawn*: the earlier $\le T+3$ reading was an artifact of comparing the depth to the un-reduced fiber budget $|U|$, and the $O_T(\log k)$ law of Prop. 3.5 admits no k-uniform linear bound. The $T = 9$ measured maximum (11) exceeds the leading-term formula (9.2) because the measurement counts oriented fibers (both j and $-j$) while the derivation is $\pm$-reduced; the discrepancy is the reflection duplicate, not a mechanism failure, and it is resolved by the $\pm$-reduced model.

T	k	measured max (oriented)	$\pm$-reduced formula
8	2	9	8.5
9	3	11	9.2
10	3	10	10.1

The calibration carries its provenance explicitly: the per-fiber cut ratios ($\approx$ 120/960/2,160), the medians, and the maxima of Table 3 are the program's three-rung cascade record from the $T = 8/9/10$ measurement runs; those runs predate the present artifact bundle, whose committed cascade scripts cover the committed cells only (depths median 2, maximum 4; §3.6). The three-rung record is therefore *calibration*, not proof input, and the derivation of the model rests on the committed machinery (Lemmas 3.1 and 8.1) alone. The measured maxima are oriented-fiber counts and exceed the $\pm$-reduced formula by the reflection duplicate (Lemma 3.6); three rungs cannot distinguish a real growth of $\bar{\gamma}(T)$ from this duplicate or from the junction fatness, but the $\pm$-reduced model removes the duplicate as a source of discrepancy.

3.4 The two-regime structure: the measured γ, the reflection fiber, and the chain class

The zone-resolved measurement of the cut quality γ over the committed cells (764 admissible fiber pairs; five cells; canonical signs; the identical seed reproducing the zoo random-40) splits into two regimes, and the split is sharp at every cell.

The reading is threefold. *(i)* The non-resonant zone satisfies H2m with enormous margin: the median pair is an instant kill—the actual intersection is *smaller* than the random model, which is the lattice sparsity—and the worst mid-zone excess over all five cells is $\gamma = 40.6$ (a single pair at $T = 7$, $p = 29$); a uniform $\gamma_0 \approx 40$ covers every committed mid-zone pair, with 90th percentiles in 14–422. *(ii)* The γ tail is entirely a $\bar{\lambda} \le 4$ phenomenon. *(iii)* The tail's dominant component is proved structure, not noise: at $T = 8$-critical every small-zone $\gamma \in [205, 819]$ is a $\lambda = -1$ fiber with survival fraction exactly 1 (the random baseline is tiny because the solution sets there are nearly rigid), and the extreme zoo values ($\le 2.09 \times 10^4$) occur at pairs with $|\mathrm{Sol}_j| \le 4$—nearly-rigid size

Table 4: Zone-resolved cut quality γ (survivors against the random baseline) over the committed cells: pair counts, median, and maximum per zone. Mid zone: $\bar{\lambda} > 4$ and $\bar{\lambda} < p - 4$; small zone: $\bar{\lambda} \leq 4$. The tail (up to 2.09×10^4) lives entirely in the small zone; its dominant component at $T = 8$-critical is the $\lambda = -1$ reflection fiber of Lemma 3.6 (survival fraction exactly 1).

	mid zone ($\bar{\lambda} > 4$)			small zone ($\bar{\lambda} \leq 4$)		
cell	n	median	max	n	median	max
$T = 7$, $p = 17$, top-10	40	0.0	0.0	68	0.0	204.7
$T = 7$, $p = 29$, top-10	120	0.0	40.6	70	0.0	508.1
$T = 8$-crit., $p = 17$, top-10	4	0.0	0.0	11	409.4	818.8
$T = 8$ zoo, $p = 17$, top-10	40	0.0	10.7	70	0.0	193.3
$T = 8$ zoo, $p = 17$, random-40	128	0.0	17.6	213	0.0	2.09×10^4

tuples where one or two survivors outweigh a minuscule baseline. Cascade depths re-confirmed on the same scope: medians 2 everywhere, maximum 4 (zoo random-40).

Lemma S. 3.6 (the free reflection fiber)**.** *The map $x \mapsto -x$ on $\mathbb{Z}_{p^2}$ preserves every bad set $B_w = \{x : \|wx\|_N \leq c\}$ and maps the fiber F_j onto F_{p-j}. Hence, for every unit system, the covering condition at fiber j is equivalent to the reflected covering condition at fiber $p - j$; in particular, for every anchor solution $a \in \mathrm{Sol}_0$ with drift μ, the reflected placement covers fiber $-j_0$ with the same drift: the fiber with dilation $\lambda \equiv -1$ has survival fraction exactly 1 and is never a cut.*

Proof. $\|w(-x)\|_N = \|wx\|_N$, so $B_w = -B_w$. For $x = j+pt \in F_j$, $-x \equiv (p-j)+p(p-1-t) \in F_{p-j}$: the involution $t \mapsto p - 1 - t$ carries the footprint $B_u \cap F_j$ onto $B_u \cap F_{p-j}$. Reflections carry arcs to arcs (an AP with canonical difference d reflects to an AP with the same canonical difference), sizes transfer (the three-tier arc law is even in a_0), and the whole fiber covering reflects to a fiber covering; the anchor's drift is a property of the composite configuration, which reflects with it. □

Measured confirmation: survival fraction exactly 1 at $\bar{\lambda} = 1$ in *every* family at *every* committed cell (840/840 for the zoo chain family $(1, 1, 1, 1)$; 6/6, 12/12, 24/24 at $T = 8$-critical). The resonance budget H2r is then derived modulo the one fitted constant: the free fiber contributes the 1, the $\bar{\lambda} \leq \Lambda_0$ dilations among the fiber set contribute $\leq 2\Lambda_0|U|/(p-1) \approx 2\Lambda_0(T-2)/T$, and $\Lambda_0 \approx 4$ is where the measured cuts turn on—data-chosen, and disclosed as such.

The case class whose structure is exactly known is the all-interval chain family $(1, \ldots, 1)$—the number-one zoo family by placement mass, with 123,240 solutions at $p = 17$ over all 32 size tuples. For it, the dilate-sparsity reduces to lattice-box counting between junction boxes, with the load-bearing invariant the overlap-sum identity.

Lemma S. 3.7 (overlap-sum identity)**.** *For the all-interval family and any size tuple s with $\Sigma s = p + \mathrm{ov}$: every covering placement, written in its sorted cyclic order σ, has the overlap vector $o_i = s_{\sigma(i)} - (a_{\sigma(i+1)} - a_{\sigma(i)})$ with $\sum_i o_i = \mathrm{ov}$ exactly, $o_i \leq s_{\sigma(i)}$, and every inter-start gap $\leq s_{\max}$ (the reach bound: the point before the next start is covered by some arc starting at or before the current one, whose reach is $\leq s_{\max}$ past it).*

Lemma S. 3.8 (chain-box parameterization)**.** $\mathrm{Sol}(s)$ *is contained in the union over the $(m-1)!$ cyclic orders σ of the affine-unimodular images of the boxes $\{o \in \mathbb{Z}^m : \sum o = \mathrm{ov},\ s_{\sigma(i)} - s_{\max} \leq o_i \leq s_{\sigma(i)}\}$; on the $o \geq 0$ stratum the parameterization is exact up to start ties, with count $\sum_\sigma \#\{o \geq 0 : \sum o = \mathrm{ov},\ o_i \leq s_{\sigma(i)}\}$. Consequently $|\mathrm{Sol}| \approx \mathrm{ov}^{m-1}$—the volume law with exponent $m - 1 = T - 4$ and constant derived from the overlap budget,* $\mathrm{ov} \approx (T-6)k$.

Both lemmas are machine-checked on the full solution set: 0 violations of either bound in all 123,240 solutions; the pure-chain ($o \geq 0$) stratum is 120,960 of them (98.2%), the nested

excess 2,280 (1.8%); the count 131,520 of (order, o) pairs against 123,240 placements is the tie/degenerate factor; the top size tuple $(5,5,5,5,5)$ gives 10,080 pairs against the committed maximum 8,880.

Theorem S. 3.9 (chain-class dilate reduction)**.** *For the all-interval family, the two-fiber survivor set at dilation λ is contained in*

$$\bigcup_{\sigma,\sigma'} \{o \in \mathrm{Box}_\sigma : \ \lambda\, \tilde{U}_{\sigma,\sigma'}\, o + \tilde{v} \in \mathrm{Box}'_{\sigma'} \pmod p\},$$

where $\tilde{U} = A_{\sigma'}^{-1} A_\sigma$ is integer-unimodular (A_σ the lower-triangular chain matrix) and the boxes have sides $\le 2s_{\max}$. Substituting $u = \tilde{U}o$ (whose box has sides $\le 2(m-1)\, s_{\max}$, the normalizer arc pinned), each term is bounded by Lemma 3.3 applied per coordinate:

$$\mathrm{survivors}(\lambda) \ \le \ ((m-1)!)^2 \Big(1 + \frac{4(m-1)\, s_{\max} + 1}{b_{\min}(\lambda,\, 2s_{\max})}\Big)^m.$$

Consequently the chain-class per-fiber cut is k-free and polynomial in $b_{\min}$: it exceeds any fixed γ once $b_{\min}(\lambda, 2s_{\max}) \ge B(T)$, and the fibers failing this are $\le B(4s_{\max}+1)$ in number—a vanishing fraction of U as k grows. Pulling the σ'-sum condition back through the dilation gives one affine congruence $\lambda\,(w_{\sigma,\sigma'}^\top o) \equiv c(\sigma,\sigma')$ per order pair; for matched sizes it degenerates to the o-independent congruence $\lambda\,\mathrm{ov} \equiv c$—either every anchor candidate satisfies it or none does: the instant-kill dichotomy, measured at $\bar\lambda \in \{2,4\}$ (survival 0) against the partial resonance at $\bar\lambda = 3$ (survival 120/840) in the zoo chain pair data.

Proof. Step 1 (the parameterization): with $a_1 = 0$ the sorted cyclic order σ of the starts has $\sigma(1) = 1$; the gaps $g_t = x_{t+1} - x_t$ partition $[0,p)$, so $\sum g_t = p$ and the overlap vector $o_t = s_{\sigma(t)} - g_t$ sums to ov (Lemma 3.7). For the reach bound: the point $y = x_{t+1} - 1$ is covered by some arc $[x_i, x_i + s_i)$ with $x_i \le x_t$ (no start lies in (x_t, x_{t+1})), whence $x_{t+1} \le x_i + s_i \le x_t + s_{\max}$. Conversely, given (σ, o) in the box with $\sum o = \mathrm{ov}$, the chain recursion $x_{t+1} = x_t + s_{\sigma(t)} - o_t$ from $x_1 = 0$ reconstructs the starts by a lower-triangular integer map A_σ with determinant ± 1. *Step 2* (the dilate pullback): a surviving pair (a, a') with $a \in \mathrm{Sol}_0(s)$ and $a' = \lambda a + \tau \in \mathrm{Sol}_j(s')$ writes $a = A_\sigma o + b_\sigma$ and $a' = A_{\sigma'} o' + b_{\sigma'}$, so $o' = \lambda \tilde{U} o + \tilde{v}$ with $\tilde{U} = A_{\sigma'}^{-1} A_\sigma \in \mathrm{GL}_m(\mathbb{Z})$. *Step 3* (the box count): the substitution $u = \tilde{U}o$ is a bijection of $\mathbb{Z}^m$ carrying the coordinate box (sides $\le s_{\max}$) into the box of sides $\le 2m\, s_{\max}$; dropping the sum-slice (enlarging), the count factors per coordinate and each factor obeys Lemma 3.3: $\le |I''|/b_{\min}(\lambda, s'_{\max}) + 1$. Multiplying over the m coordinates and the $((m-1)!)^2$ order pairs gives the displayed bound. *Step 4* (the sum congruence): for actual solutions Lemma 3.7 gives $\sum o' = \mathrm{ov}'$ as integers; pulling through the affine expression, $\lambda\,(w^\top o) \equiv \mathrm{ov}' - 1^\top \tilde{v} \pmod p$ with $w = \tilde{U}^\top 1$, a necessary congruence per order pair; o-independent for matched sizes. □

What the theorem does and does not give. It gives the exact structural account of the measured H2 phenomenology for the dominant family: the instant kills (the sum-congruence failure), the confinement of the resonance to small $\bar\lambda$ (the $b_{\min}$ regime), the k-freeness of the cut, and the volume law that fixes the depth formula's leading constant. It does *not* give the sharp uniform $\gamma_0 = O(1)$ with clean constants—the order factors $((m-1)!)^2$, $4(m-1)\, s_{\max}$ are loose by orders of magnitude at $T = 8$, where the measured mid-zone cut is an instant kill and the bound gives ≈ 1; the route to sharpness is to count *on the slice* (keep $\sum o = \mathrm{ov}$ in the lattice count—it is load-bearing; dropping it is what costs the constants), a geometry-of-numbers lemma in the genre of successive minima. And it does not cover the distinct-difference families, whose junction structure is not a chain: that is H1g, the subject of the next subsection.

3.5 The volume law at $T = 8$: the census, renormalization, and the strand census

H1g is the volume law with uniform margin, $|\mathrm{Sol}(f, \mathrm{sz})| \le c_1 \,\mathrm{ov}^{m-1}$ for the $\pm$-distinct families, with $c_1 \,\Gamma \le \big(T/(T-6)\big)^{m-1} e^{-\eta}$ for an $\eta > 0$ uniform in (T, k, f, sz)—the margin clause that keeps Proposition 2.2's denominator positive (the $\mathrm{Hq}_{\pm}$ constant Γ is now included). The bare-c_1 form below is the measured content; the margin clause is the open content. It was attacked at the smallest rung where the pattern is visible, $T = 8$, by full census where the cell admits it: $p = 17$ ($k = 2$, $\epsilon = 1$; all 840 families, all 32 size tuples, ov $\in [3, 8]$; per-family totals median 2,981, 90th percentile 9,633; per-(family, size) maximum 876 at $(5, 5, 5, 5, 5)$, ov $= 8$); $p = 19$ ($k = 2$, $\epsilon = 3$; all 1,680 families; admissible 960, exactly the pool saturation of Table 2 reproduced by an independent code path; totals median 102, 90th percentile 645, maximum 801; per-size maximum 108 at ov $= 6$); and $p = 29$ ($k = 3$, $\epsilon = 5$; *targeted*: every cascade-containing set, the doubling-path quadruples, the anchors, and a 120-set random sample—138 of the 715 difference sets, 710 families of the 17,160 total, a scope disclosed and not claimed beyond; 20/138 sets admissible, consistent with the full-scope 0.133; per-size maximum 186 at $(8, 8, 8, 8, 8)$, ov $= 11$). Every decision procedure was validated against an independent direct brute force (plain p^4 enumeration, no cascade, no pruning) before any number was cited: 9/9 keys match exactly across the three primes, including the chain controls 8,880/4,680/29,760 and zero keys.

Lemma S. 3.10 (renormalization invariance)**.** *Let $D = \{1\} \cup S$ be the difference multiset of a census family (S a $\pm$-distinct $(m-1)$-set), and for $\delta \in D$ let* $\mathrm{renorm}_\delta(S) = \{\mathrm{canon}(d\,\delta^{-1} \bmod p) : d \in D,\ d \ne \delta\}$, $\mathrm{canon}(x) = \min(x, p - x)$. *Then the total covering count summed over all size assignments is equal for S and* $\mathrm{renorm}_\delta(S)$*:* $\sum_{\mathrm{sz}} |\mathrm{Sol}(S, \mathrm{sz})| = \sum_{\mathrm{sz}} |\mathrm{Sol}(\mathrm{renorm}_\delta(S), \mathrm{sz})|$.

Proof. The covering condition is invariant under the global dilation $\varphi(x) = \delta^{-1}x$ of $\mathbb{Z}_p$ (a bijection carrying an arc $a + d\,[0, s)$ onto the arc $\delta^{-1}a + (d\,\delta^{-1})[0, s)$). Apply φ to a covering placement of the family S with size assignment $(s_d)_{d \in D}$: the image is a covering placement of the family with difference multiset $\delta^{-1}D$, in which the arc that had difference δ now has difference 1—it is the normalizer of the transformed system. Re-anchor at that arc's start (covering is translation-invariant) and canonicalize each difference $d \mapsto \mathrm{canon}(d)$ (an arc with difference $-d'$ is the reflection, $x \mapsto -x$ composed with a translation, of an arc with difference d'; reflections preserve coverings). The result is a census-normal-form covering placement of $\mathrm{renorm}_\delta(S)$ with the sizes carried along. Summing over all size tuples removes the bookkeeping dependence on which arc received which size (the admissible size-tuple set is permutation-closed), and the map is a bijection at each size assignment. □

Verified exactly: at $p = 17$ the 35 difference sets partition into 7 renormalization orbits (each of size 5, one per choice of $\delta \in D$), and the 7 volume classes are precisely these orbits; at $p = 19$ the 70 sets form 14 orbits (8 positive, 6 zero), and the 9 volume classes are exactly the unions of orbits. The volume is therefore a function of the dilation class of the difference multiset alone—not of the family key. This cuts the family space 5-to-1 in general and identifies the correct equivalence for any future classification: at $T = 8$ the census object is not the $\binom{D-2}{m-1}$ sets but their renormalization orbits. It is a dilation-invariance statement in the spirit of the class reduction at prime moduli (Lemma 4.6) and the census law of Conjecture 6.7: it identifies the equivalence; it does not count. For the cascade the consequence is load-bearing: the two-fiber cut at dilation λ compares Sol_0 with a dilate of Sol_j, and Lemma 3.10 says the volumes involved are dilation-class invariants—the cut law's input volumes are orbit functions, and H1g needs constants uniform over orbits, not over keys.

The reading of Table 5 is fourfold. *(i)* The uniform bound with $c_1 = 1$ holds at every measured prime, with margin growing in k (and tightening with the ϵ window at fixed k). *(ii)* The distinct class is uniformly *thinner* than the chain class at equal ov, by a factor growing from 10× to 160×. *(iii)* The direction of risk for H1g is therefore not a fat counterexample family but proof

Table 5: The volume law at $T = 8$: the worst $\pm$-distinct renormalization orbit against the chain benchmark, at the top size tuple. Normalized: the maximum divided by ov^4. The uniform bound $|\mathrm{Sol}| \le \mathrm{ov}^4$ (that is, $c_1 = 1$) holds at every census prime with margin $4.7\times$ /$12\times$ /$79\times$, *growing in* k; the distinct class sits $10\times$ /$43\times$ /$160\times$ below the chain class at equal ov. The $p = 29$ top orbit is identified within the targeted scope; a family outside it would have to exceed ov^4 outright to change the law's direction, and the admissibility collapse argues against it.

p	top distinct orbit	max @ ov	normalized	chain @ tuple	ratio
17	$\{1,2,3,4,8\}$	876 @ 8	0.214	8,880	0.099
19	$\{1,2,4,5,8\}$	108 @ 6	0.083	4,680	0.023
29	$\{1,2,4,7,8\}$	186 @ 11	0.013	29,760	0.0063

coverage—the opposite of an obstruction. *(iv)* The caveats: three primes, two k-values, both ϵ classes at $k = 2$ but only one at $k = 3$; the $p = 29$ top orbit rests on the targeted scope; the measurements fix the law's direction and margin, not its proof.

Proposition S. 3.11 (the threshold form of H1g)**.** *The per-coordinate bound* $|\mathrm{Sol}(f, \mathrm{sz})| \le p^{m-1}$ *is trivial at every cell of the measured family (the moving-start space is* $\mathbb{Z}_p^{m-1}$*), and* $p/\mathrm{ov} \to T/(T-6)$ *because* $\mathrm{ov} = (T-6)k + O(1)$ *against* $p = Tk + \epsilon$*, so the trivial bound realizes*

$$c_1^{\mathrm{triv}} = \big(T/(T-6) + o(1)\big)^{m-1}$$

in the normalization of the volume law—exactly the value at which the depth denominator $(m-1)\log(T/(T-6)) - \log c_1$ *of Theorem 2.1 vanishes. At* $m - 1 = T - 4$ *the threshold is* 256*,* 243*,* 244.1 *at* $T = 8/9/10$*: near-constant across rungs, so neither the cascade's task nor the volume law's softens at higher* T*.*

Proof. The first inequality is the per-coordinate count: a solution is a choice of the $m-1$ moving starts, each in $\mathbb{Z}_p$. The second is the size window of Lemma 8.1: the $m = T-3$ arcs of the two-value window give $\mathrm{ov} = \Sigma s - p = (T-6)k + O(1)$ against $p = Tk + \epsilon$. Dividing, $|\mathrm{Sol}| \le (p/\mathrm{ov})^{m-1}\mathrm{ov}^{m-1}$, which is the volume law with $c_1 = (p/\mathrm{ov})^{m-1}$; substituting this c_1 in the depth formula of Theorem 2.1 zeroes its denominator, so the trivial bound purchases no cascade depth at all. □

The threshold form restates what H1g must purchase: not the free bound $|\mathrm{Sol}| \le p^{m-1}$ but the *margin* over it, growing in k. Measured, the margins below the trivial box are $95\times$/$1{,}207\times$/$3{,}803\times$ at $p = 17/19/29$ (top distinct family at its top size tuple: $17^4/876$, $19^4/108$, $29^4/186$—the maxima of Table 5); the committed margins over ov^4 ($4.7\times$ /$12\times$ /$79\times$) are the same fact in volume-law units. The form also unifies the program's constant-quality obligations under one numerical criterion: every volume constant must beat $\big(T/(T-6)\big)^{T-4} \approx 250$. Every measured constant clears it—the chain class's 2.0–3.6 by roughly two orders of magnitude, the distinct class by the factors above—and no proved constant yet does (the chain constants of Theorem 3.9 are loose by orders of magnitude). The sharp-constants question of §3.4 and H1g are one criterion stated twice.

The mechanism of the top class was probed on the full solution sets. Every solution needs all five arcs (essential-arc pattern $(1,1,1,1,1)$ in 876/876): no spectator redundancy. The overlap-budget identity $\sum_{i<j} O_{ij} = \mathrm{ov} + \sum_x \binom{o(x)}{2}$ holds exactly (0 violations), with pairwise overlaps small (mean 0.5–1.5, maximum 2–3) and under the spacing caps with room: no pair carries the covering; the arrangement is collective. Coordinate projections of the solution set at the probed families sit far from full ($[16,17,14,12]$ for the top family; $[9,9,7,8]$ for the minimum class)—per-family statistics that initially read as a $T = 7$-style pinning law persisting at the zoo cell; the exhaustive key-level check of the character-system paragraph below shows that reading is an artifact, not a law. And the block geometry exhibits the mechanism: the interval covers

$[0, s_1)$; the $d = 2$ arc covers every other point of a window of span $2s - 1$; the $d = 4$ arc every fourth point of a span $4(s - 1)$; the $d = 8$ arc is the antipodal two/three-block sampler; the circle closes by interleaving these against the interval and each other. The top class is the *binary cascade* $\{1, 2, 4, 8\}$ plus a fifth difference—at $p = 17$ the orbit swallows every fifth value $\{3, 5, 6, 7\}$; at $p = 19$ the fifth is 5; at $p = 29$ it is 7 (the observed pattern "fifth $= \epsilon + 2$" holds at the three primes measured; three points, flagged as such, not a law). At $p = 29$, where 2 is a primitive root, the canonicalized doubling orbit visits every difference and only cascade-containing quadruples survive among the path quadruples: resonance structure is necessary for admissibility at $k = 3$, consistent with the pool collapse.

The character system: the corrected coordinates. The projection reading above was tested as a law and is false in both naive forms. Call the *character marginals* of a solution set the projection sizes along the coordinate functionals of the moving-start space. The two natural invariance readings—(V′) the multiset of the four singleton marginals is a renormalization-orbit invariant, and (V″) the same for the six pairwise-difference marginals—are both false, each caught by machine check (V′: profile multisets differ across orbit members at both full primes; V″: 0/32 σ-matched size tuples agree at $p = 19$). The mechanism was pinned by an explicit construction of the renormalization bijection (at $p = 19$, $S = (2, 4, 5, 8)$, $\delta = 5$, with image $S' = (3, 4, 6, 8)$): the image of each moving slot is δ^{-1} times a source slot coordinate *minus the anchor coordinate* a_δ, plus a size-dependent constant, and the new normalizer is $\delta^{-1}a_\delta$—every moving slot is coupled to the anchor. Consequently the image's slot singletons are the source's anchor pairs, and the image's pairwise differences (in which the anchor cancels) are the source's non-anchor singletons and pairs: neither subsystem is closed, and the union is.

Lemma S. 3.12 (the extended character system)**.** *Let $X = \{e_i\} \cup \{e_i - e_j : i < j\}$ on the moving-start space $\mathbb{Z}_p^{m-1}$, so $|X| = (m - 1) + \binom{m-1}{2}$ (ten functionals at $T = 8$). Under the renormalization bijection of Lemma 3.10, each functional of X maps to a nonzero scalar multiple of a functional of X plus a size-dependent constant. Consequently the multiset of marginal sizes $\{|\pi_\chi(\mathrm{Sol}(S, \mathrm{sz}))| : \chi \in X\}$, aggregated over the permutation-closed set of size tuples, is a renormalization-orbit invariant.*

Proof. The closure statement is the anchor-mixing computation just recorded, verified symbolically and then end-to-end at $p = 19$ for $S = (2, 4, 5, 8)$: all 801 images of $\mathrm{Sol}(S, \mathrm{sz})$ land in $\mathrm{Sol}(S', \mathrm{sz} \circ \sigma)$, injectively, at every size tuple, with per-key counts matching under the explicit σ (0 mismatches)—which also confirms the per-assignment-bijection step of Lemma 3.10's proof at a stronger level than the committed record (totals were verified; assignments now are). The invariance follows: each character's marginal size is carried to another character's marginal size up to the size permutation, so the aggregate multiset is constant along the orbit. Verified on the full top orbits at both primes—$p = 17$: the orbit of $(2, 3, 4, 8)$, five members, 9,633 solutions per set; $p = 19$: the orbit of $(2, 4, 5, 8)$, five members, 801 per set—with the aggregate ten-functional multisets identical across members at both primes; and at $k = 3$, the two admissible $p = 29$ probe sets $(2, 4, 7, 8)$ and $(3, 4, 8, 13)$ form one orbit ($\delta = 7$) with equal totals 1,311. □

Two consequences. First, the coordinate system for the census is now fixed: any class-level (orbit-level) law must be stated on X—not on absolute starts, not on pairwise phases alone—so the symmetry analysis of the census object is complete (Lemma 3.10 fixes the equivalence, and Lemma 3.12 the coordinates in which a law can be stated). Second, the *marginal route into the strand census is measured dead* at the zoo cell. Across 1,678 admissible keys (exhaustive at $p = 17$ and $p = 19$, targeted at $p = 29$; every key inside the committed census scope matches it, 1,654/1,654, 0 mismatches), the projection envelope over all keys and all ten characters is p itself at both $k = 2$ primes and 27/29 at $k = 3$ (fullness fractions 1.000/1.000/0.931); a sufficient marginal law on X— $|\pi_\chi| \le \gamma_0 \,\mathrm{ov}$ with $\gamma_0^{m-1} < \left(T/(T-6)\right)^{m-1}$—would imply the volume law, since Sol is contained in the product of its singleton marginals, but it requires the fullness fraction

to fall below $\gamma_0(T-6)/T$ asymptotically (below 75% already at $\gamma_0 = 3$), and no downward trend is in evidence at the three committed primes. The top-family profiles are non-monotone in (k, ϵ) ($[12, 17, 17, 16] \to [4, 4, 9, 8] \to [9, 7, 10, 27]$; the free character 27 at $p = 29$ is the $d = 8$ antipodal sampler of the block geometry above): family- and window-dependent statistics, not a law. The only invariant sparsity is *joint*: the median density of Sol inside its own marginal box is 1.5%/7.4%/6.9% at the three primes—the pointwise no-holes condition along I, which is what the strand language below encodes.

The strand census: where H1g now lives. The reduction runs in three steps, each elementary. *Step 1* (the interval spine): the complement of the normalizer interval is the interval $I = [s_1, p)$ of length $\ell = p - s_1 \approx (T-2)k$; every covering of $\mathbb{Z}_p$ by the interval plus the $m-1$ moving APs restricts to a covering of I by the I-restrictions of the moving APs (plus the two boundary points at the wrap), and the restriction map is injective up to the boundary bookkeeping, so an upper bound on the I-restricted count bounds |Sol|. The moving arcs' total size is $(m-1) \cdot 2k \approx (T-3) \cdot 2k$ against $\ell \approx (T-2)k$: a near-exact cover with slack ov. *Step 2* (the strands): the I-restriction of an AP with step d is a union of *strands*—maximal runs of consecutive t whose points $a + td$ stay inside I; within a strand the points sit d apart. Covering consecutive points of I requires strands of different steps to interleave; a pair (d, d') interleaves only at compatible phases, and the phase-compatibility condition per pair is a three-distance-type statement about $\lambda = d'/d$—exactly the Lemma 5.16 genre, on the pair lines $td - ud' \equiv \delta$ that already govern the pairwise overlaps. *Step 3* (the arrangement form): every covering induces a cyclic *arrangement*—the maximal runs of I and, for each run, the subset of strands carrying it, with the handoffs between runs subject to the pair-phase conditions—and the volume decomposes as $|\mathrm{Sol}| \le \sum_{\text{arrangements}}$(phase freedom of the arrangement), with boundary corrections.

What is missing for a proof is the uniform count: bound the number of arrangements (the handoff patterns) and the phase freedom per arrangement, uniformly over the resonance classes of the difference set—equivalently, by Lemma 3.10, over the renormalization orbits. The chain class is the degenerate resonance class where all steps coincide and the strand picture collapses to the telescoping of Lemmas 3.7–3.8; the binary cascade is the pure ratio-2 class; a general orbit mixes ratio classes. The three families of necessary conditions already available—the overlap budget, the pair spacing caps, and the last-arc fit count—do *not* discriminate: random placements typically satisfy the budget (the expected total pairwise overlap ≈ 12 exceeds ov at $p = 17$). The discriminating structure is the pointwise no-holes condition along I, which is what the strand language encodes. The reduction is to a named, bounded combinatorial object, the *strand census*; the injectivity of the restriction map and the boundary corrections are machine-consistent at the census cells but are not proved in general, and the strand census is therefore a *plausible* reduction of H1g, not an established equivalence. The work is class-by-class (per resonance type), and uniformity across classes is itself a lemma—the precise shape of the remaining distance.

The transfer-matrix route: the third route, closed by theorem. The third candidate route—a transfer-matrix or automaton model of the joint density under the arc-covering constraint—was examined as a proof attempt under a pre-committed success criterion: a bound on the whole-cascade survival fraction, survivors $\le \rho^{\mathrm{depth}} \cdot$ initial mass, with $\rho \le 1 - \epsilon$ for some $\epsilon > 0$ independent of k. Two specifications were fixed in advance, as the route's design demanded. *The state space*: the covering constraint at fiber j is membership $c(j) \in \mathrm{Sol}(\mathrm{sz}(j))$—a function of the placement state alone; the cascade is non-consuming (the alive-set semantics is a pure intersection), so the constraint is Markovian in the natural state $(\mathbb{Z}_p)^{m-1}$, and the theorem below is invariant under state augmentation in any case, so no resolution of this specification could change the verdict. *The comparison target*: the whole-cascade fraction, with both factors' k-dependence acknowledged—the initial mass is $|\mathrm{Sol}_0|$, and the per-step ratios inherit the k-scaling of the forbidden sets.

Lemma S. 3.13 (the monomial dichotomy of the cascade, single-anchor scope)**.** *Fix the rung, the prime, and any sequence of fibers $j_1, \ldots, j_D$ from the uncovered set with solution sets $\mathrm{Sol}_t = \mathrm{Sol}(\mathrm{sz}(j_t))$, and let σ_t be the affine bijection between consecutive fiber placements of Lemma 3.1. Then:* (i) *every constrained transition $P_{\mathrm{Sol}_{t+1}}\,\sigma_t\,P_{\mathrm{Sol}_t}$ is a partial permutation matrix (at most one nonzero entry, equal to* 1*, per row and per column), and so is every product along the chain; its spectrum is* $\{0\}$ *(the directed paths) together with the roots of unity of the surviving cycle lengths.* (ii) *The affine maps compose to the identity around any closed fiber loop (each re-expresses the fixed drift μ; the dilation ratios telescope), so the cyclic transfer product is a diagonal* 0/1 *matrix whose trace is the number of placements surviving the loop—the cascade's own survivor count—and whose spectral radius is* 1 *if that count is positive and* 0 *otherwise.* (iii) *Consequently every constrained transfer matrix obtainable in the framework—open chain, cyclic, homogenized single pair, or state-augmented—has spectral radius exactly* 0 *or* 1*: never an interior value, at any k.*

Proof. (i) σ_t is a bijection; composing with coordinate projections leaves at most one nonzero per row and per column, and partial permutations are closed under products. The digraph of a partial permutation has in- and out-degree at most 1, hence decomposes into directed paths and cycles; paths give nilpotent blocks, and a cycle of length ℓ gives a cyclic permutation block whose spectrum is the ℓ-th roots of unity. *(ii)* Each σ is the change of drift coordinates $c(j') = (j'/j)\,c(j) + c^0(j') - (j'/j)\,c^0(j)$ (Lemma 3.1); composing around a loop telescopes the dilation ratios to 1 and the translations to 0, and deletion only zeroes diagonal entries. *(iii)* Augmentation preserves determinism (the register update is a function of the state), so the augmented constrained matrix is again a partial permutation, and parts (i)–(ii) apply verbatim. □

Scope. The lemma is true but narrow: it closes only the *single-anchor* automaton—the operator that tracks one anchor placement along the drift. It does *not* close the general strand census, whose natural transfer operator scans I position by position with local coverage states and *branches* (a state records which strands are active at the current position, and several are), so its constrained operator is not a partial permutation and may have interior spectral radius—as the 2×2 example above ($\rho \approx 1.122$) shows. The lemma therefore does *not* establish that "the toolkit is provably not the right shape" in general; it establishes the weaker, correct statement that *this* single-anchor transfer operator consumes H1g rather than proving it, every spectral quantity in its framework being 0 or 1. The general strand-census automaton is the open route; the lemma's scope is the single-anchor special case, and the abstract and §2 state the general version only as the open obligation.

The machine record runs at three levels. *Generic*: 1,500/1,500 random constrained systems (random bijections with random deletion sets) satisfy the structural dichotomy—the partial-permutation premise, and acyclic if and only if nilpotent—and 12,000 exact trace fingerprints $\mathrm{tr}(Q^t) = \sum_{\ell \mid t} \ell\, c_\ell$ (over 300 systems, $t \le 40$, integer-exact) match the cycle inventory with zero mismatches. *Committed objects*: six committed families (the zoo chain and top-distinct orbits at $p = 17$ and $p = 19$; the two committed admissible $p = 29$ sets), 66 homogenized pair systems—the route's own model, iterating a single pair map $x \mapsto \lambda x + \tau$ under the deletion set Sol_j—of which exactly the 6 reflection pairs ($\bar{\lambda} = 1$) have $\rho = 1$ and all 60 others have $\rho = 0$; the loop-composition identity of part (ii) holds algebraically and numerically (500 random points per family); the automaton's per-step survivor counts equal the committed μ-space cascade convention at every step (the chain family's depth 2 reproduces the committed zoo record); and the end-to-end ground truth—$[\sigma(x) \in \mathrm{Sol}_j]$ if and only if the direct fiber-covering test accepts $\sigma(x)$—holds on 2,800/2,800 sampled checks, with the committed volume margins reproduced exactly (top $|\mathrm{Sol}| = 876/108/186$, margins $95\times$ / $1{,}207\times$ / $3{,}803\times$ over the trivial box). *The reflection fiber in automaton language*: the two-fiber survival fraction at $\lambda = -1$ is exactly 1 at all six families across all three primes, and the homogenized reflection sub-shift closes the whole

of Sol_j into 2-cycles—survival flat forever: Lemma 3.6 as a statement about the instrument itself, the deletion at the free fiber being empty.

What the instrument measures at the committed cells is equally sharp. The iterated-pair survival counts collapse as $[840, 120, 0]$, $[224, 1, 0]$, $[4{,}680, 240, 0]$—piecewise extinction within three iterates, the intermediate values set by maximal lattice runs of solution points along the drift orbits (runs of length at most 2 everywhere; extinction confirmed at every horizon to 12)—the three-distance genre of Lemma 5.16, not geometric decay with an interior rate. And the k-scaling that the obligations need (the margins $95\times$ / $1{,}207\times$ / $3{,}803\times$) lives in $|\mathrm{Sol}|/p^{m-1}$, in the census counts, and in no spectral quantity: the spectral outputs are binary at every prime, with no k-trend because there is nothing to trend.

Nor does any reformulation of the instrument escape the dichotomy. A coarse-grained or weighted automaton (states as classes, class multiplicities as weights) can carry an interior spectral radius numerically, but for non-negative matrices the radius dominates every directed cycle's geometric mean, so $\rho \le 1-\epsilon$ would force uniform decay of the weight products along drift orbits—and the weights are census counts, so this is the margin itself entering as an input rather than an output; the converse fails besides (the 2×2 witness $\left(\begin{smallmatrix} 0 & 0.9 \\ 0.9 & 0.4 \end{smallmatrix}\right)$: every directed cycle has geometric mean ≤ 0.9 yet $\rho \approx 1.122$—branching resurrects mass), so a second external input, a branching control, would be needed as well. Averaged or randomized operators collapse to the independent-extension baseline, whose deficit is the γ-correlation—the cascade model's own deepest unproved input. Every door leads through an external uniform-in-k estimate of exactly the margin type; none generates one. In automaton language the whole-cascade survival fraction is the product of the consecutive conditional ratios $r_t = |S_t|/|S_{t-1}|$, each bounded by $\gamma_0\,|\mathrm{Sol}_{j_t}|/p^{m-1}$ under H2m with the thinness of $|\mathrm{Sol}|$ supplied by H1g: under its own model the route's output is Proposition 3.5's recursion, and the uniform deficit it owes is the hypothesis stack it was commissioned to prove. The instrument is the census in different notation.

The verdict against the pre-committed trichotomy of outcomes: the success criterion ($\rho \le 1-\epsilon$) is unattainable by theorem—there is no interior spectral radius to bound, in any state formulation, at any k; and the failure mode is not $\rho \to 1$ as $k \to \infty$ but $\rho \in \{0, 1\}$ as an identity of the lossless dynamics. The type-mismatch reading is thereby upgraded from observation to mechanism *for the single-anchor operator*: bijections preserve measure exactly (the monomial structure), deletion filters without dissipating (partial permutations carry spectrum in $\{0\}$ together with the unit circle), and the loop composition is the identity (the cyclic instrument measures alive-or-dead, not decay). Margins require dissipation; losslessness forbids generated dissipation *for this operator*. The reformulation is bounded by its own fidelity *at the single-anchor level*. The scope caveat is load-bearing: the general strand-census automaton branches and is not covered by the lemma, so the stronger claims "the toolkit is provably not the right shape" and "bounded by its own fidelity" in general are *not* established here and are withdrawn from the abstract and §2; what is established is the single-anchor dichotomy. Lemma 3.13 is the program's third k-invariance theorem—after the spacing lemma's k-cancellation and the character system's closed coordinates—and, like them, a closure-type statement: the single-anchor toolkit's signature holds even for the instrument built to break it.

3.6 The verification record of this section

Every number above is either an exhaustive census with two independent decision procedures, or a sample with its size and interval, or a proved statement with its machine check; the records, in the order of appearance: the pool censuses (the pruned-cascade and the flattened-popcount procedures agree exactly at $p = 17/19/23$; an independent pure-enumeration brute force agrees on 21/21 sampled tuples across $p = 17/19/23/31$, covering both admissible and inadmissible branches; the $T = 9$ stride samples carry full decisions, and the $m = 6$ code path shares the cascade code brute-force-verified at four $m = 5$ primes, with the external consistency check the agreement of the projected pools with the independently measured pool scale); the zone-

resolved γ table (764 pairs, five committed cells, the identical seed reproducing the committed family selections); the chain-class machine checks (123,240 solutions, 0 violations of the sum identity or the reach bound; the count reconciliations 131,520 against 123,240 and 10,080 against 8,880); the reflection fiber (survival exactly 1 at every $\bar{\lambda} = 1$ pair, every family, every cell); the renormalization-orbit law (machine-checked set-by-set at $p = 17$ and $p = 19$); the volume census and its ground truth (9/9 brute-force keys; the admissibility reproductions 840/840 and 960/1,680; the order-independence of volume across all 24 orderings); the essential-arc, budget-identity, and cap checks on the probed families; and the cascade depths at the committed cells (medians 2, maximum 4); and the character-system record of §3.5 (the rigidity census over 1,678 admissible keys with every key inside the committed census scope matching it exactly, 1,654/1,654, 0 mismatches; the per-assignment renormalization bijection, 801/801 images injective with per-key counts matching under the explicit σ; the falsifications of V' and V''; the character-system aggregate invariance on both full top orbits; the chain controls 8,880/4,680/29,760 reproduced by the same code path). The transfer-matrix route's record: the generic dichotomy and trace fingerprints (1,500 systems, 12,000 comparisons, zero mismatches); the 66 homogenized pair systems on the six committed families with the closed-form orbit inventories verified by full traversal and the finite-horizon survival counts verified by direct simulation (zero mismatches); the loop-composition identity (algebraic and numeric, 500 points per family); the per-step equality of the automaton's survivor counts with the committed μ-space cascade convention (zero mismatches, all steps, all families; the committed chain-family depth reproduced); the end-to-end fiber-covering ground truth (2,800/2,800); the reflection pairs (survival fraction exactly 1, all six families, all three primes); and the reproduction of the committed volume margins (876/108/186 against $95\times$ /1,207 $\times$ /3,803$\times$). The three-rung depth record of Table 3 is calibration from the earlier cascade runs and is labeled as such wherever it is used.